5.1 Unit description

Introduction

In this unit the study of electrode potentials builds on the study of redox in Unit 2, including the concept of oxidation number and the use of redox half equations.

Students will study further chemistry related to redox and transition metals.

The organic chemistry section of this unit focuses on arenes and organic nitrogen compounds such as amines, amides, amino acids and proteins. Students are expected to use the knowledge and understanding of organic chemistry that they have gained over the whole International Advanced Level in Chemistry when covering the organic synthesis section.

This unit draws on all other units within the International Advanced Level in Chemistry and students are expected to use their prior knowledge when learning about these areas. Students will again encounter ideas of isomerism, bond polarity and bond enthalpy, reagents and reaction conditions, reaction types and mechanisms. Students are also expected to use formulae and balance equations, and calculate chemical quantities.

How chemists work

The study of chemical cells provides an opportunity to illustrate the impact on scientific thinking when it emerges that ideas developed in different contexts can be shown to be related to a major explanatory principle. In this unit, cell emfs and equilibrium constants are shown to be related to the fundamental criterion for the feasibility of a chemical reaction: the total entropy change.

The explanatory power of the energy-level model for electronic structures is further illustrated by showing how it can help to account for the existence and properties of transition metals. In this context there are opportunities to show the limitations of the models used at this level and to indicate the need for more sophisticated explanations.

Study of the structure of benzene is another example that shows how scientific models develop in response to new evidence. This links to further investigations of the models that chemists use to describe the mechanisms of organic reactions.

The study of catalysts touches on a 'frontier' area for current chemical research and development which is of theoretical and practical importance. This provides an opportunity to show how the scientific community reports and validates new knowledge.

Students have further opportunities to carry out quantitative analysis, to interpret complex data and assess the outcomes in terms of the principles of valid measurement. The topic of organic synthesis illustrates a selection of the techniques that chemists have developed to carry out reactions and purify products efficiently and safely.

Core practicals

The following specification points are core practicals within this unit that students should complete:

5.3.1d

5.3.1g

5.3.2g

5.3.2j

5.4.1d

5.4.1e

5.4.2b

5.4.2d

5.4.2i

5.4.3f

These practicals may appear in the written examination for Unit 5.

Use of examples

Examples in practicals

Where 'e.g.' follows a type of experiment in the specification students are not expected to have carried out that specific experiment. However they should be able to use data from that or similar experiments.

For instance in this unit, *5.3.1h i Application of redox equilibria*, the specification states:

demonstrate an understanding of the procedures of the redox titrations below (i and ii) and carry out a redox titration with one:

- i *potassium manganate(VII), e.g. the estimation of iron in iron tablets*.

Students will be expected to have carried out a redox titration with potassium manganate(VII), but they may or may not have done this to estimate the amount of iron in iron tablets.

In the unit test students could be given experimental data for a potassium manganate(VII) titration, in any context, and be expected to analyse and evaluate this data.

Examples in unit content

Where 'e.g.' follows a concept students are not expected to have been taught the particular example given in the specification. They should be able to illustrate their answer with an example of their choice.

For instance in this unit*, 5.4.2h Organic nitrogen compounds:*

amines, amides, amino acids and proteins, the specification states:

comment on the physical properties of polyamides and the solubility in water of the addition polymer poly(ethenol) in terms of hydrogen bonding, e.g. soluble laundry bags or liquid detergent capsules (liquitabs).

Students will be expected to comment on the physical properties of polyamides and the solubility of poly(ethenol) in terms of hydrogen bonding, but they may or may not have looked at soluble laundry bags or liquitabs.

In the unit test students could be asked to comment on the physical properties of this polyamide and the solubility of the addition polymer in terms of hydrogen bonding. This could be in the context of soluble laundry bags, or in another completely different context.